

TRM Token Tool Experiments (SmolLM-135M Sudoku Calls)

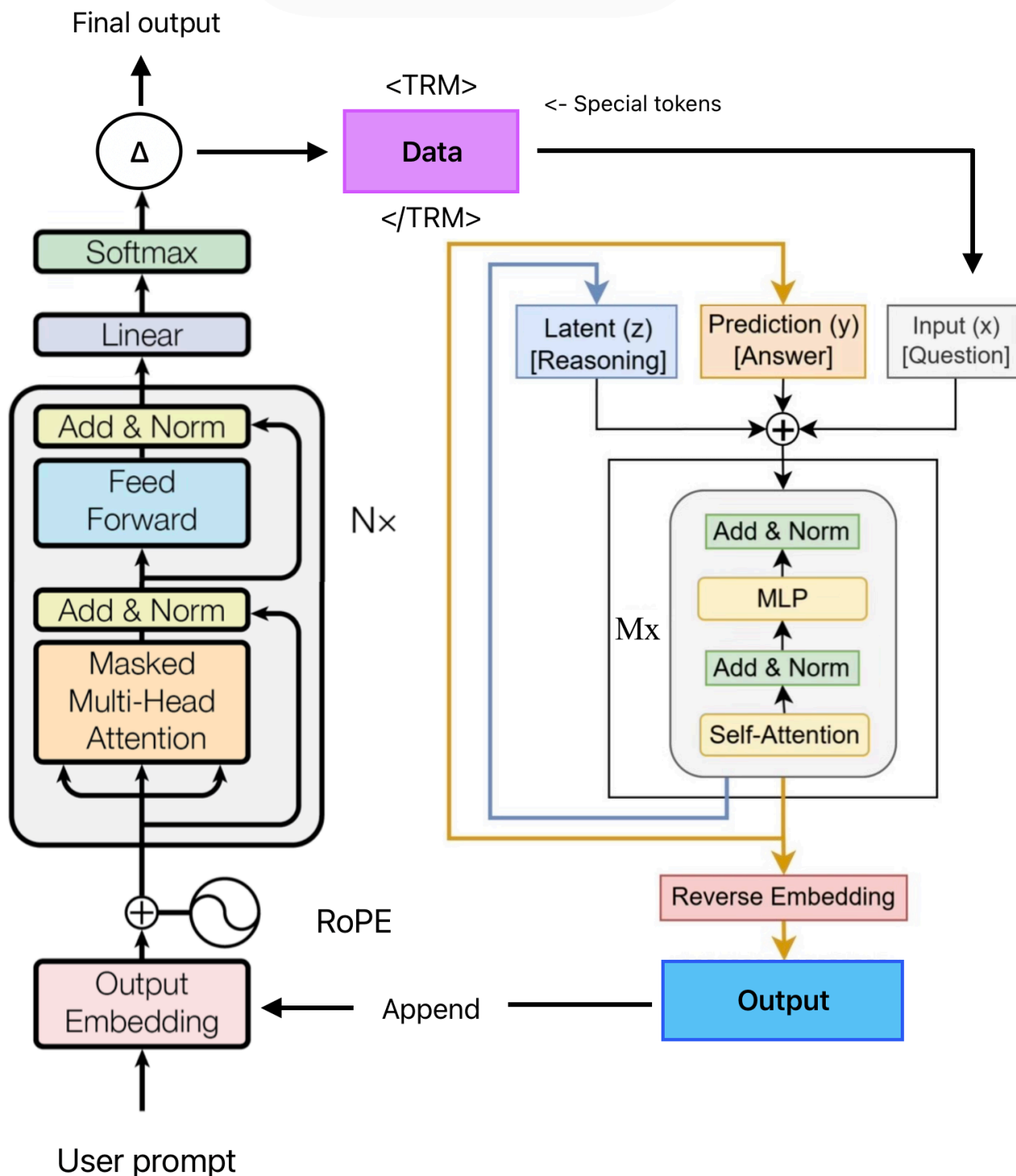


Figure 1: TRM token call: the model emits a Sudoku payload between special tags.

Overview

SmolLM-135M is fine-tuned to emit `<TRM-sudoku> ... </TRM-sudoku>` calls so an external Python solver can handle Sudoku. The dataset mixes Sudoku prompts with chat and general text to reduce catastrophic forgetting. Training uses L2 regularization toward the base checkpoint and bfloat16 compute.

Experiment summary

Experiment	Dataset	Train rows	Mix (S/C/T)	Steps	L2	Simple wiki delta	OOD prompt
Sudoku only	trm_sudoku_calls	v1 5000	1.0/0.0/0.0	5000	0	+0.8800	n/a
Mix v1	trm_sudoku_mix	v1 8000	0.3/0.2/0.5	8000	1e-5	-0.0089	0/3 @ 256
Mix v2	trm_sudoku_mix	v2 10000	0.4/0.2/0.4	10000	1e-5	+0.0251	0/3 @ 256
Mix v3	trm_sudoku_mix	v3 10000	0.45/0.2/0.35	10000	1e-5	+0.0345	0/3 @ 256
Mix v4 (current)	trm_sudoku_mix	v4 10000	0.5/0.2/0.3	10000	1e-5	+0.0364	3/3 @ 512

Loss over time (mix v4)

Loss curve is evaluated on 10 batches (batch_size=2, max_rows=256) from the v4 train mix at each 1k checkpoint.

Step	Train-mix loss
1000	1.0471
2000	0.5969
3000	0.6157
4000	0.6111
5000	0.9905
6000	0.8445
7000	0.6101
8000	0.4612
9000	0.6068
10000	0.7350

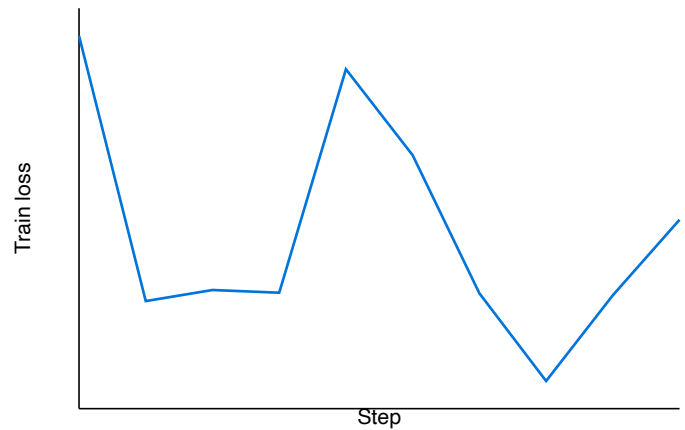


Figure 2: Mix v4 loss snapshots vs checkpoint step.

Performance summary

Mix runs keep general loss near the base model while improving OOD tool usage.

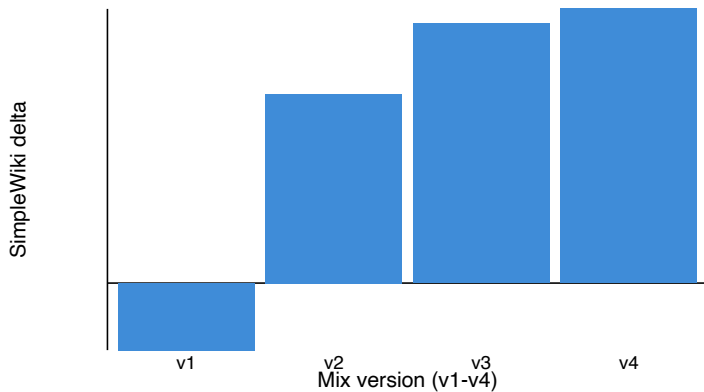


Figure 3: Simple wiki delta (fine - base), mix runs only.

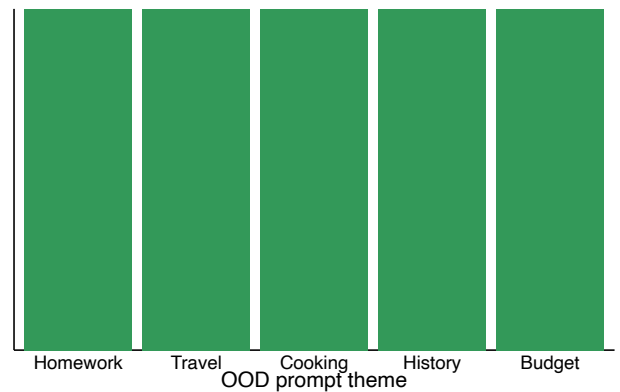


Figure 4: OOD prompt success (10 runs each).

Evaluation protocol

- General loss: Run `eval_general_loss.py` on `simple_wiki` (20 batches, seq_len 256).
- OOD prompts: 5 long mixed-topic prompts from `trm_token_tool/ood_eval/prompts.json`, 10 runs each with `Run_eval_ood_sudoku.py`; success means the parsed puzzle matches the input and the solver matches the generator solution.